

সেচবন্ধু: WEB-BASED BANGLA AI AGENT

Voice-Interactive Project Assistant for Smart Irrigation Rover

A separate website page where users speak in Bangla, view live transcripts, receive project-based AI answers, and hear Bangla voice replies.

/ai-agent Page

Groq AI + Next.js API

Bangla Voice Reply

AI AGENT INNOVATION

The website includes a Bangla web-based AI assistant named “Sechbondhu”. Users can ask questions by voice, see a live Bangla transcript, and receive short Bangla answers about the Smart Irrigation Rover.

The agent is not a simple chatbot. It is connected with project knowledge, a secure backend route, Groq AI, and Bangla audio playback.

WHY WE ADDED IT

Many users may not understand technical details such as ESP32, sensors, dry-zone logic, and smart irrigation. The AI agent works like a Bangla project guide inside the website. It helps farmers, visitors, and evaluators ask questions naturally and understand the system quickly.

USER INTERFACE FEATURES

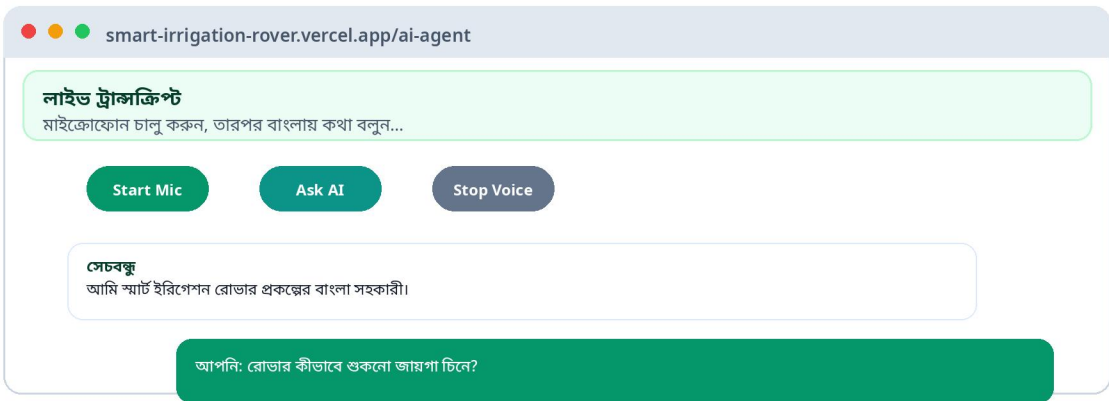
- Separate Ai-Agent page
- Start Mic and Stop Mic buttons
- Live Bangla transcript box
- Ask AI button
- Chat-style Q&A display
- Bangla audio reply
- Stop Voice button
- Project-based AI behavior

CURRENT STATUS

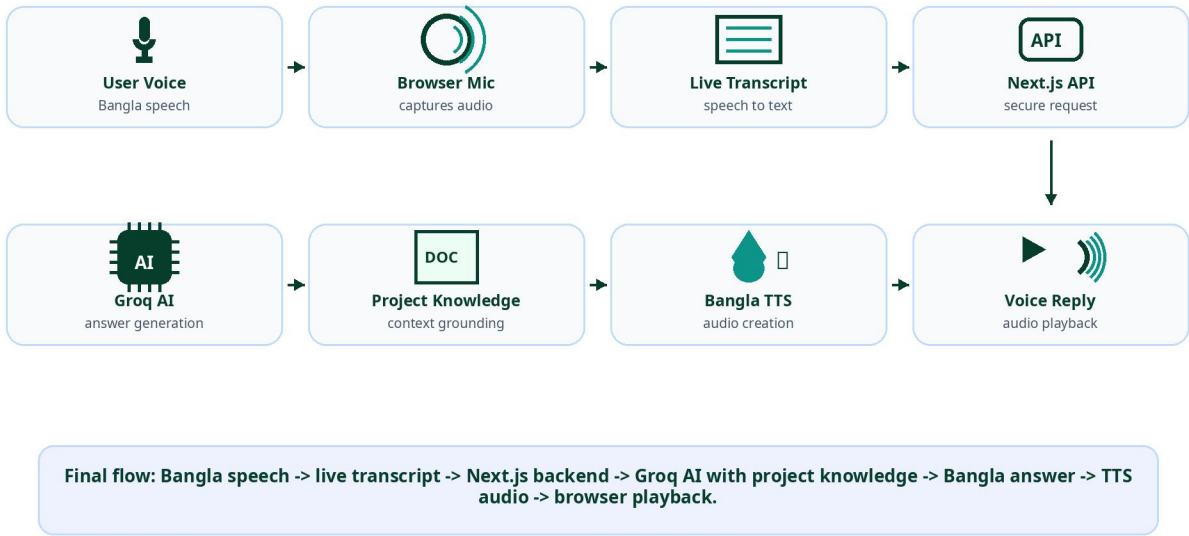
Implemented: separate AI page, navbar link, browser microphone input, live transcript, Groq AI response, project knowledge, Bangla TTS route, and audio playback.

Future: connect voice commands with ESP32 for pump control and sensor queries.

WEBSITE PAGE DESIGN



COMPLETE AI AGENT FLOW



STEP-BY-STEP IMPLEMENTATION LOGIC

- Page Route**
Created /ai-agent as a separate page and added navbar link.
- VoiceAgent Component**
Handles microphone, transcript, chat UI, AI call, and audio playback.
- AI API Route**
POST /api/robot-ai sends user question to Groq securely.
- Project Knowledge**
lib/project-knowledge.ts grounds answers in rover details.
- Bangla TTS Route**
POST /api/bangla-tts creates playable Bangla audio.
- Voice Output**
Audio is played in the browser as the AI reply.

MAIN FILES USED

- app/ai-agent/page.tsx**
Separate AI Agent page.
- components/voice-agent.tsx**
Mic, transcript, chat, AI call, audio playback.
- app/api/robot-ai/route.ts**
Groq AI backend route.
- app/api/bangla-tts/route.ts**
Bangla audio generation route.
- lib/project-knowledge.ts**
Project facts used by AI.

SYSTEM PROMPT LOGIC

- Name: Sechbondhu
- Answer only in simple Bangla
- Avoid fake project claims
- Use project knowledge context
- Keep answers short for voice
- Mention future integration when needed

DEMO EXPLANATION

- Open the Ai-Agent page.
- Click Start Mic.
- Ask a question in Bangla.
- Watch the live transcript.
- Click Ask AI.
- Groq generates a project-based Bangla answer.
- The website plays the answer as Bangla voice.

BENEFITS AND FUTURE

- Benefits:
- Makes the project interactive
 - Helps Bangla-speaking users
 - Improves presentation quality
 - Works as a project guide
- Future Integration:
- ESP32 pump control by voice
 - Sensor data query by voice
 - Real-time rover status through AI